

Physics-Informed Residual Diffusion for OD-Conditioned Trajectory Generation

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- **Phase A (pipeline)**: strict, leakage-free dt-fixed protocol (split-first, train-only products).
- **Phase B (what we built)**: physics-informed **residual diffusion** + inference-time CFG; strong OD alignment and micro realism (best-of- K).
- **Phase C (what breaks)**: **Destination Gravity / topological collapse** — detours are missing even with CFG.
- **Phase D (pivot)**: reframe as **trip-level decision + execution**; follow SOTA hierarchical planning.

Task Definition

- Task: **KnownDestination** OD-conditioned trajectory generation (grid space, dt-fixed).
- Windows: history $H=8$, horizon $F=12$ (dt-fixed $\Delta t = 30\text{s}$).
- Key challenge: **multi-modal trip-level decision + continuous execution**.
- Goals:
 - **Coverage**: best-of- K (oracle) ADE/FDE, distribution metrics (Fréchet/DTW).
 - **Physical realism**: macroscopic motion (MSD curve, Rog).

Outline

Phase A: Pipeline

Phase B: Residual Diffusion (What Works)

Phase C: Failure Mode (Destination Gravity)

Phase D: Pivot (Decision + Execution)

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Phase A (from raw data to strict pipeline)

Relation to previous stage: starting from raw Shenzhen taxi GPS records, we need a reproducible pipeline before any model comparison.

Phase A objective: eliminate ambiguity and leakage (dt, split, nav_field source, normalization).

Phase A outputs:

- dt-fixed dataset ($\Delta t = 30\text{s}$) and train/val/test splits
- train-only products: data_stats.json (normalizer) + nav_field.npz
- sanity checks (split overlap, metadata source, coordinate range, alignment)

Phase A: Strict Data Contract (KISS)

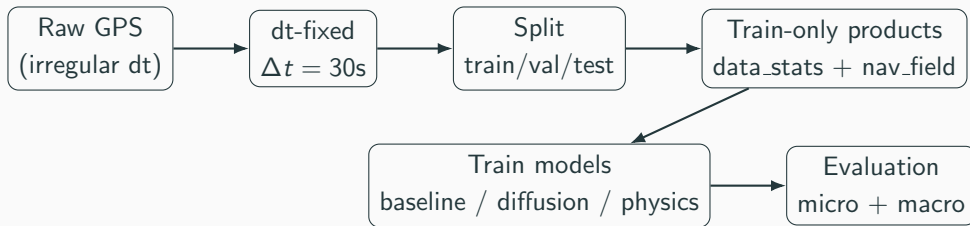


Figure 1: *

Split-first, train-only products: prevents nav_field/statistics leakage into val/test.

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Our Method: Physics-Informed Residual Diffusion

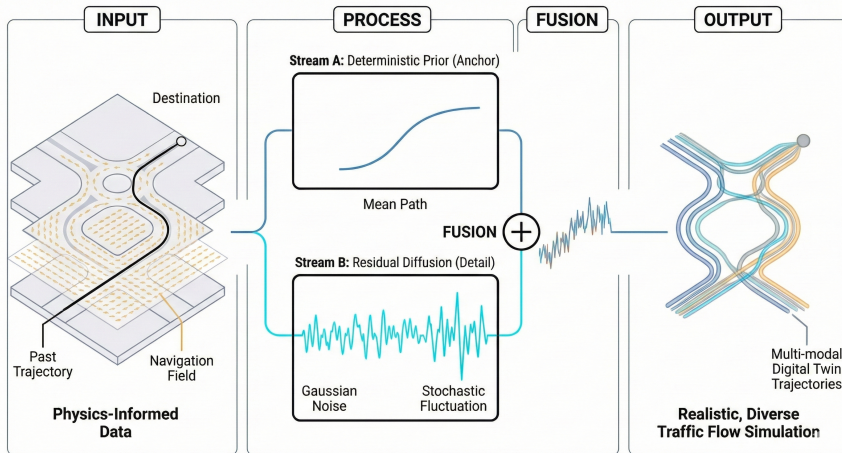


Figure 2: *

Macro: OD alignment in geographic space (Density, N=10k)

Reading guide: GT density on the left; model density on the right (with GT contour). High sample size reveals corridor-level structure.

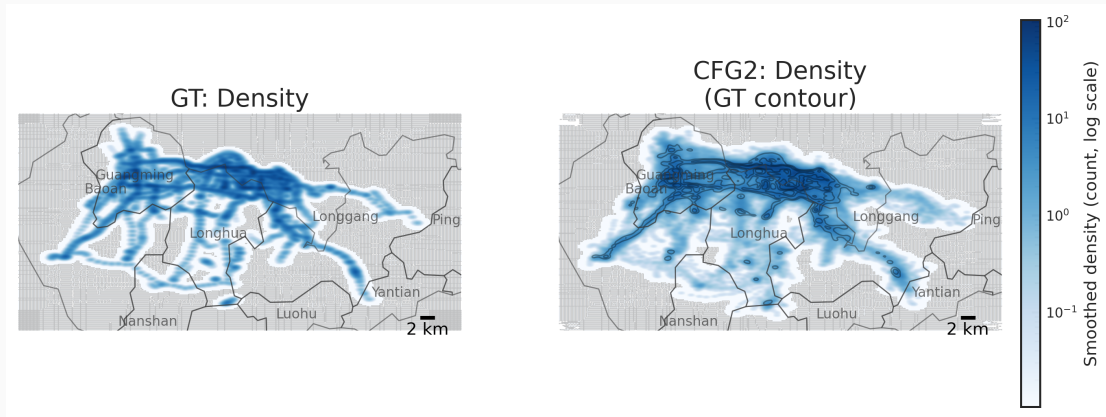
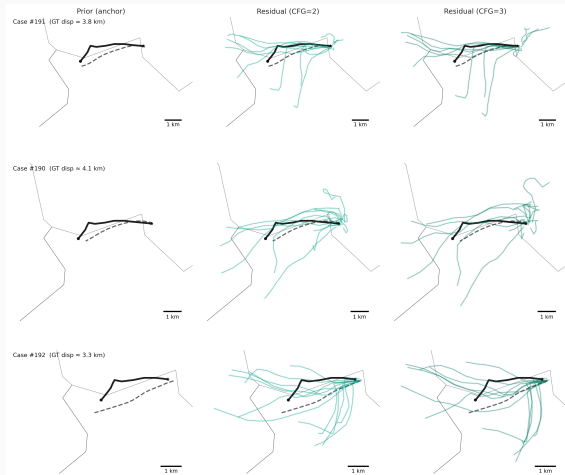


Figure 3: *

Micro: Multi-modal local execution (best-of- K)

Reading guide: Left = deterministic prior (anchor), Mid/Right = residual diffusion samples under CFG (K spaghetti). Residual adds local realism and diversity.



Physical Statistics (Validity): Speed/Turn/Accel + MSD

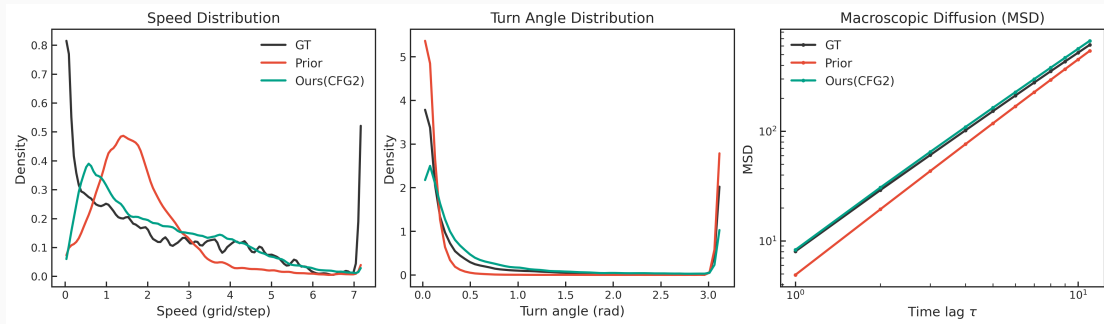


Figure 5: *

Statistical consistency improves (JSD vs GT), but this alone does not guarantee correct trip-level topology.

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The Defect: Destination Gravity / Premature Convergence

Observation: regardless of Prior vs CFG2/CFG3, many predicted trajectories are “pulled” straight to the destination; GT often detours (e.g., to reach a highway entrance).

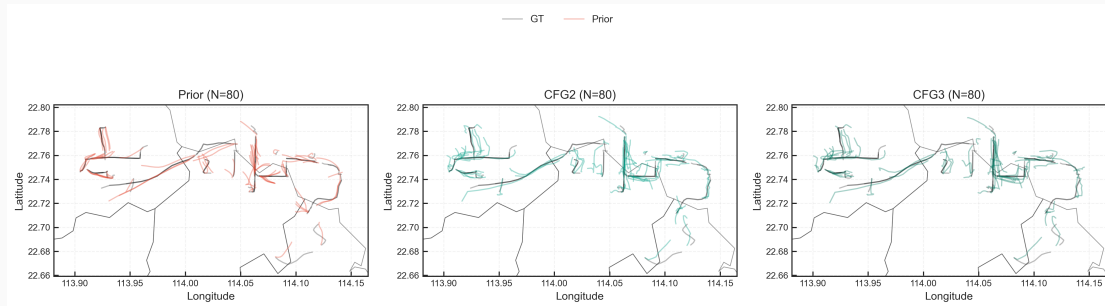


Figure 6: *

Trajectory overlays (Prior vs CFG2 vs CFG3): topology collapses towards a straight-to-goal path.

Why CFG cannot create detours (first-principles view)

- Conditioning on destination is like a **potential well** at the goal: without intermediate barriers/anchors, trajectories follow the steepest descent (“straight-to-goal”).
- CFG scales this gradient at inference: it can change **tail/jitter/violations**, but it does **not** introduce new low-frequency topological modes.
- Oracle diagnostics (workstation runs): even with GT mid-way waypoint, turn-angle distribution does not recover $\Rightarrow$ detour modes are largely absent from the model support.

Evidence snippet (validity metrics, dt-fixed)

Example (N=10k, K=1): statistical consistency improves, but topology can still collapse.

Method	JSD_{Turn}	$\text{JSD}_{\text{Speed}}$	$\text{JSD}_{\text{Accel}}$	$\text{DCV}_{\text{Speed}}$	$\text{DCV}_{\text{Accel}}$
Prior	0.0956	0.1423	0.2537	0.128%	0.002%
Ours (CFG2)	0.0503	0.0727	0.0531	0.580%	0.733%

- JSD improves: generated trajectories look more realistic statistically.
- But detour requires **low-frequency trip-level decisions**, not just local fluctuations.

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Reframing: trip-level decision + continuous execution

- **Decision (macro):** choose a route topology / a few waypoints (highway entrance, bridge, etc.).
- **Execution (micro):** generate a smooth, feasible trajectory that follows the chosen plan.
- Our current end-to-end generator mixes them
⇒ easy shortcut: “straight-to-goal + jitter”.

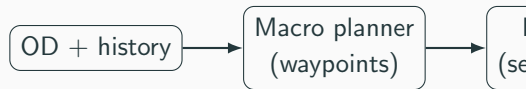


Figure 7: *

Two-stage view: decision then execution.

How SOTA addresses Destination Gravity

- **Graph/Map-based diffusion** (e.g., road graph generation): topology is constrained by the road network, detours become natural turns at intersections.
- **Hierarchical / coarse-to-fine diffusion**: first generate a few waypoints (macro), then generate detailed paths between them (micro).

Family	Topology validity	Data requirement	Engineering cost
Graph/Map-based	strong	high (road graph)	high
Hierarchical (waypoints)	strong detours	low (GPS only)	medium

Our next step: hierarchical decision + residual execution (KISS)

- **Macro stage (decision)**: predict a few waypoints (multi-modal) given OD + history.
- **Micro stage (execution)**: residual diffusion conditioned on waypoints to generate smooth segment trajectories.
- **Gate (avoid wrong turns)**: validate with detour-sensitive metrics (turn-angle distribution + low-frequency detour) + DCV.
- **Fallback**: only if hierarchical is insufficient, consider road graph / map constraints.

References (selected)

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